

Identities needed to prove the matrix exponential in the line just above Murray, Li and Sastry equation (2.14)

$$\hat{q}^2 = a a^T - \|a\|^2 I$$

$$\hat{q}^3 = -\|a\|^2 \hat{a}$$

$$\hat{\omega}^2 = \omega \omega^T - I \quad \text{since} \quad \|\omega\| = 1.$$

$$\hat{\omega}^3 = -\hat{\omega}$$

$$\hat{\omega}^4 = \hat{\omega}^3 \hat{\omega} = -\hat{\omega} \hat{\omega} = I - \omega \omega^T$$

$$\hat{\omega}^5 = \hat{\omega}^3 \hat{\omega}^2 = -\hat{\omega} \hat{\omega}^2 = -\hat{\omega}^3 = \hat{\omega}$$

$$\hat{\omega}^6 = \hat{\omega}^5 \hat{\omega} = \hat{\omega}^2 = \omega \omega^T - I$$

$$\hat{\omega}^7 = \hat{\omega}^5 \hat{\omega}^2 = \hat{\omega} \hat{\omega}^2 = -\hat{\omega}$$

$$\hat{\omega}^8 = \hat{\omega}^7 \hat{\omega} = -\hat{\omega}^2 = I - \omega \omega^T$$